

Technology Stack

Date	10 July2026
Team ID	XXXXXX
Project Name	Smart Lender – Loan Prediction System
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	HTML, CSS, Bootstrap	To build a responsive and user-friendly web interface for entering loan details and displaying prediction results

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	Python (Flask)	Provides fast processing and integrates easily with the Machine Learning model.
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	CSV Dataset & Pickle (.pkl) Model	Stores training data and the trained Machine Learning model used for prediction.
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	Local Flask Server / GitHub	Used for project development, testing, and version sharing.
5	Version Control & CI/CD <i>(e.g., GitHub, GitLab, Jenkins)</i>	Git & GitHub	Enables version control, collaboration, and secure project backup

6	Third-Party APIs / Other Tools	Scikit-learn, Pandas, NumPy	Used for data preprocessing, model training, and loan eligibility prediction.
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